Sustainable "Rendezvous": A Festival Systems Challenge

End-Semester Presentation — Modules 3.1–3.5 (Computational Results)

Sustainability Task Force

Department of Chemical Engineering

Course: CLL782 – Process Optimization
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Problem Statement

The Challenge

"Rendezvous" — IIT Delhi's annual cultural festival — draws **40,000+** visitors over 3 days onto a 163-acre campus not designed for that load.

How do we plan the festival's sustainability infrastructure — waste bins, collection vehicles, and water stations — optimally?

Sub-Problems (Modules 3.1–3.5)

- **3.1** Environmental load modelling
- **3.2** **3-type** dustbin placement (FLP)
- **3.3** Waste logistics with **vehicle segregation**
- **3.4** Water refill station planning (P-Median)
- **3.5** Integrated multi-objective optimization

What Makes This Different

- Real **geospatial data** — not textbook distances
- **OSM walking network** + Dijkstra
- Gravity-model footfall estimation
- 3-type waste & vehicle segregation
- LP relaxation bounds for every module

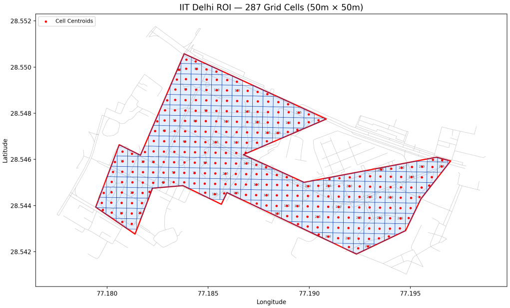
Outline

From Google Earth to MILP: The Geospatial Pipeline

Google Earth KML Polygon
18 vertices
→
UTM Grid Generation
50m × 50m
→
OSM Network Download
413 nodes, 1078 edges
→
Dijkstra's Shortest Path
287 × 287 matrix

Why Not Just Use Euclidean Distance?

- Euclidean **underestimates by ∼30%** (buildings, fences)
- Real paths follow **sidewalks & footpaths**
- Dijkstra on OSM = actual pedestrian reachability
- Every D_{ij} in Mods 3.2–3.5 is a **real walking distance**



287 grid cells, OSM network (blue), centroids (red) on IIT Delhi campus

Geospatial Pipeline: Step-by-Step Details

Step 1: ROI Extraction

- 18-vertex polygon from Google Earth KML
- **162.8 acres** (OAT, Nalanda, SAC, LHC, Sports Complex)
- Projected to **UTM** for metric gridding

Step 2: Grid Generation

- 50m $\times$ 50m cells in UTM projection
- Cells with <25% ROI overlap discarded
- **287 valid cells**: each is both a demand zone i and candidate site j

Step 3: OSM Walking Network

- Downloaded via `osmnx` (ROI + 200m buffer)
- **413 nodes**, **1,078 edges** (footpaths, roads)
- Each cell centroid **snapped** to nearest node

Step 4: Walking Distance Matrix

- Single-source Dijkstra from every snapped node
- 287 $\times$ 287 symmetric matrix
- Avg: **836.9m**, Max: **3,351m**
- Fallback: Euclidean $\times$ 1.3 if disconnected

Output: `walking_distance_matrix.csv` — the D_{ij} feeding **every** MILP module.

Footfall & Waste: Spatial Gravity Model

Gravity Model for Footfall Distribution

$$F_i = \frac{\sum_{v=1}^{10} w_v \, e^{-d_{iv}^2/2\sigma^2} \cdot A_i}{\sum_{k=1}^{287} \sum_v w_v \, e^{-d_{kv}^2/2\sigma^2} A_k} \times N_{total}$$

F_i = footfall in cell i (persons)
w_v = attractiveness weight of venue $v \in [0.3, 1]$
d_{iv} = haversine distance from cell i to venue v (m)
σ = Gaussian decay length-scale = 350 m
A_i = area of cell i (m^2)
N_{total} = total peak footfall = 40,000 persons

3-Type Waste per Cell (CPCB Norms)

Waste per person = 0.15 kg/visit. Split:

- Compostable: 40% = 2,400 kg/day
- Recyclable: 40% = 2,400 kg/day
- General: 20% = 1,200 kg/day

Metric	**Value**
Peak Footfall	40,000
Min Cell Footfall	2.6 persons
Max Cell Footfall	288.5 persons
Total Waste	6,000 kg/day

3.1 Environmental Load Function

Model: Total Environmental Load E (kg CO_2-eq)

$$E(N,S,A) = \underbrace{(\alpha_1 N + \alpha_2 S + \alpha_3 A)}_{\text{Base Load}} + \underbrace{(\beta_N N^{1.3} + \beta_S S^{1.2} + \beta_A A^{0.8})}_{\text{Non-linear Scaling}} + \underbrace{\gamma \frac{N^2}{S}}_{\text{Congestion}}$$

Variables: N = attendees, S = food stalls, A = activity-hours.

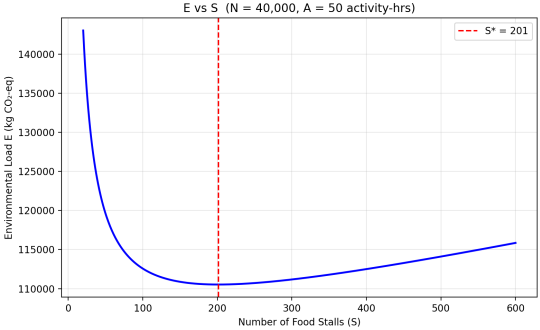
Coefficients (from literature):

$$\begin{array}{ll} \alpha_1 = 2.5 & \beta_N = 0.002 \\ \alpha_2 = 18.0 & \beta_S = 0.5 \\ \alpha_3 = 12.0 & \beta_A = 5.0 \\ = & \gamma = 0.0005 \end{array}$$

Goal: Find S^* minimising E at N=40,000, A=50.

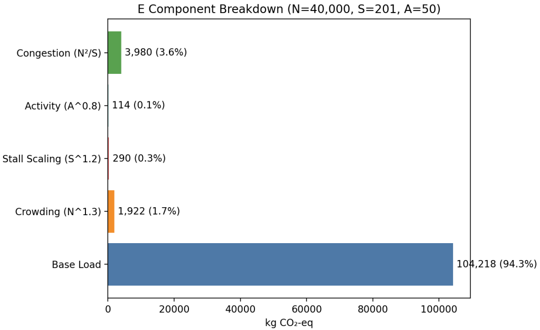
Results

$S^* = 201$ stalls; $E^* = 110{,}524$ kg CO_2-eq.
Per-capita: 2.76 kg CO_2e/person.

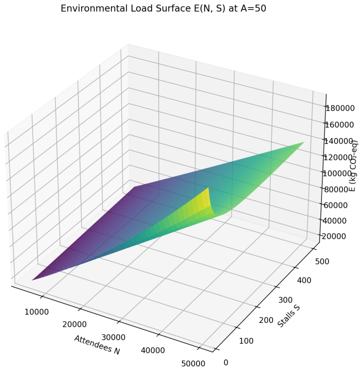


E vs S at fixed N=40,000, A=50. Minimum at $S^* \approx 201$.

3.1 Component Breakdown & 3D Surface



Component decomposition at design point



3D surface: E(N, S) at A=50

Insight: Congestion penalty dominates at low S. Base load is the largest single component at the optimum.

3.2 Formulation: 3-Type Waste-Segregated FLP

Key Change from Midsem

3 bin types $t \in \{\text{compostable}, \text{recyclable}, \text{general}\}$ with type-specific capacity, cost, and service radius.

Decision Variables

- $y_{j,t} \in \mathbb{Z}_{\geq 0}$: Number of bins of type t at site j
- $a_{i,j,t} \in [0, 1]$: Fraction of zone i's type-t waste assigned to bin j

Bin Type	Color	Capacity (kg)	Cost (Rs.)	Radius (m)
Compostable (Wet)	Green	20	10,000	200
Recyclable (Dry)	Blue	15	8,000	200
General (Inert)	Grey	10	6,000	250

3.2 MILP Formulation

Objective: Minimise Total Waste-Weighted Walking Distance

$$\min \; Z = \sum_{i=1}^{287} \sum_{j} \sum_{t \in \mathcal{T}} W_{i,t} \cdot a_{i,j,t} \cdot D_{ij}$$

Where: $W_{i,t}$ = waste of type t in zone i (kg), D_{ij} = walking distance from i to j (m, from Dijkstra), $\mathcal{T} = \{\text{compostable, recyclable, general}\}$.

Subject to:

1. **Full coverage**: $\sum_j a_{i,j,t} = 1 \;\; \forall\, i, t$ (every zone's waste assigned)
2. **Bin capacity**: $\sum_i W_{i,t} \cdot a_{i,j,t} \leq K_t \cdot y_{j,t} \;\; \forall\, j, t$ (K_t = capacity per bin of type t, kg)
3. **Service radius**: $a_{i,j,t} = 0$ if $D_{ij} > R_t$ (R_t = max walk to bin type t, m)
4. **Budget**: $\sum_j \sum_t C_t \cdot y_{j,t} \leq 35{,}00{,}000$ (C_t = unit cost of bin type t, Rs.)

Solved as MILP using **PuLP/CBC**. LP relaxation also solved for lower bound.

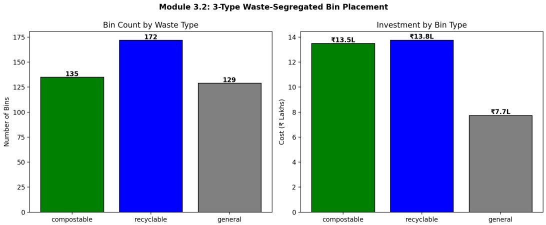
3.2 Computational Results

MILP Solution

- Compostable: 135 bins at 108 sites
- Recyclable: 172 bins at 120 sites
- General: 129 bins at 99 sites
- **Total: 436 bins**
- Cost: Rs. 35,00,000 (100% of budget)
- Objective: 93,209 (waste-m)

LP Relaxation

Objective = 0 (bins at exact demand sites).
LP lower bound: 400 bins, Rs. 32,01,440.



Bin counts and cost by waste type

3.3 Formulation: Vehicle Type Segregation

Objective: Minimise Total Daily Logistics Cost

$$\min \sum_{t\in\mathcal{T}} \sum_{i} \sum_{j\in\mathcal{F}_t} \underbrace{\left(c_t^{var} \cdot L_j + c^{hdl} + c_j^{proc}\right)}_{\text{unit cost per kg}} \cdot x_{i,j,t} + \sum_{t} c_t^{fix} \cdot N_t$$

Variables: $x_{i,j,t} \geq 0$ = kg of waste type t shipped from zone i to facility j; $N_t \in \mathbb{Z}_{\geq 0}$ = vehicles of type t deployed.
Where: c_t^{var} = Rs./km for vehicle type t; L_j = distance to facility j (km); c^{hdl} = handling cost (Rs. 1/kg); c_j^{proc} = processing cost at j (Rs./kg); c_t^{fix} = daily fixed cost per vehicle; $\mathcal{F}_t$ = facilities accepting type t.

Waste t	**Facility j**	L_j
Compostable	Biogas (cap 500 kg)	1 km
Recyclable	Okhla Recycler	11 km
General	Okhla Landfill	12 km

Vehicle	**Cap**	c_t^{fix}	c_t^{var}
Green Tipper	750 kg	600	15
Blue Tipper	750 kg	600	15
Grey Tipper	500 kg	500	18

3.3 Results: Base Scenario

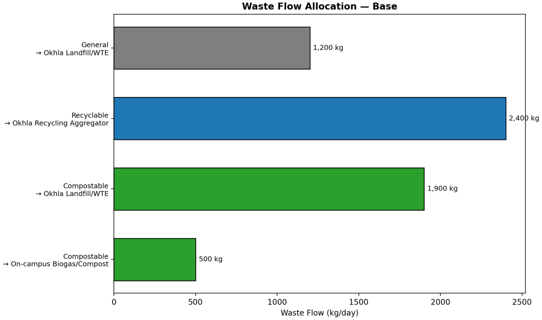
Optimal Flow Allocation

- Compostable: 500 kg → Biogas, 1,900 kg → Landfill
- Recyclable: 2,400 kg → Recycler
- General: 1,200 kg → Landfill

Cost Breakdown

- Transport + Handling: Rs. 10,10,700
- Processing: −Rs. 12,750 (recycling revenue!)
- Fleet Fixed: Rs. 3,400
- **Total: Rs. 10,01,350**

Fleet: 2 Green + 2 Blue + 2 Grey = **6 vehicles**.

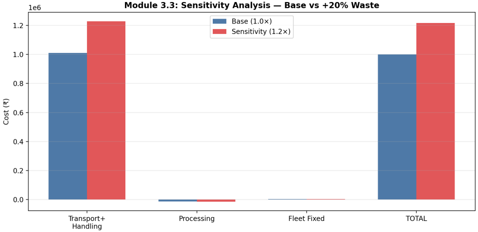


Waste flow allocation — base scenario

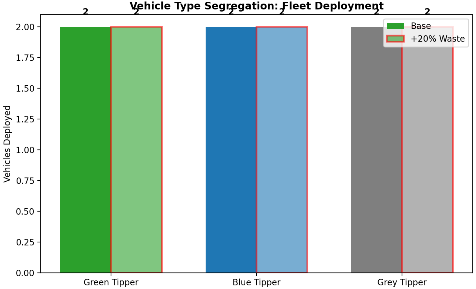
3.3 Sensitivity Analysis: +20% Waste

Key Findings

- Biogas at 500 kg cap even in base case.
- **1,900 kg compostable → landfill** (base); 2,380 at +20%.
- Cost: +21.6% (Rs. 2,16,240).
- Same fleet size (absorbed by trips).
- **Bottleneck**: composting capacity.



Cost comparison: Base vs +20% waste



Vehicle deployment by type

3.4 Capacitated P-Median: Water Refill Stations

Objective

$$\min \underbrace{\sum_j f \cdot y_j}_{\text{Installation}} + \underbrace{\sum_{i,j} F_i \cdot x_{i,j} \cdot D_{ij} \cdot c_w}_{\text{Walking inconvenience}}$$

Variables: $y_j \in \{0,1\}$: open station at j; $x_{i,j} \in [0,1]$: frac. of zone i assigned to j.

Parameters: f=Rs. 1L/station; F_i=footfall (persons); D_{ij}=walking dist. (m); c_w=Rs. 0.02/m; Q=1,000 persons/station.

Constraints: $\sum_j x_{i,j}=1\ \forall i$ (demand met); $\sum_i F_i x_{i,j} \leq Q y_j\ \forall j$ (capacity); $x_{i,j}=0$ if $D_{ij}>500$m.

Results (Greedy + LP Bound)

- **40 stations**, Rs. 40,00,000
- Avg walk: **75 m**, Max: 2,492 m
- Capacity = demand = 10,000 LPH
- **LP gap: 1.31%**

3.5 Multi-Objective Optimization

Formulation: Budget-Constrained Weighted Sum

$$\min_{B_{bin},\, B_{stn}} \quad w_1 \hat{C}_{sys} + w_2 \hat{E}_{sys} + w_3 \hat{I}_{sys} \qquad \text{s.t.}\ \ B_{bin} + B_{stn} \leq B_{total} = \text{Rs. } 85\text{L}$$

Decision Variables: B_{bin} = bin budget; B_{stn} = station budget. **Weights:** $w_1 + w_2 + w_3 = 1$, swept over simplex.
Objectives ($(\hat{\cdot})$ = min-max normalised to $[0, 1]$): C_{sys} = total economic cost (Rs.); E_{sys} = CO_2e from infrastructure + waste processing (kg); I_{sys} = walking inconvenience (person·m·Rs.). Solved via SciPy SLSQP (multi-start).

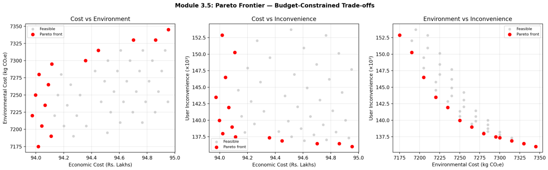
	Min Cost	**Min Inconv.**
Bins	404	535
Stations	40	40
C_{sys}	Rs. 83.9L	Rs. 95.0L
E_{sys}	6,690 CO_2e	7,345 CO_2e
I_{sys}	160,657	136,026

Recommended (Balanced)

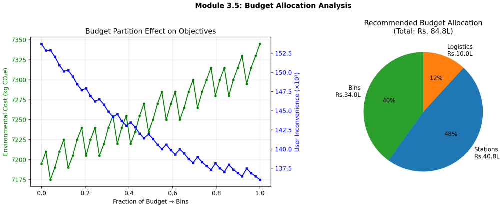
Weights: $w_1 = 0.29$, $w_2 = 0.31$, $w_3 = 0.41$

- **404 bins + 40 stations**
- Bins Rs. 34L + Stations Rs. 40.8L
- Total: Rs. 83.9L, CO_2e: 6,690 kg

3.5 Pareto Frontier & Budget Allocation



Pareto frontier: Cost vs Environment vs Inconvenience



Budget partition effects and recommended allocation

Key Insights:

- Stations always at minimum (40) — capacity constraint is tight.
- Extra bin budget yields diminishing returns on inconvenience.
- **Expanding composting capacity** would shift the Pareto frontier favourably.

Integrated Blueprint & Key Takeaways

System-Level Summary

1. **Mod 3.1**: $S^* \approx 201$ stalls; $E = 110{,}524$ kg CO_2-eq at design point.
2. **Mod 3.2**: **436 bins** (3 types) across 287 zones; full waste segregation at source.
3. **Mod 3.3**: **6 segregated vehicles** (Green/Blue/Grey); biogas bottleneck at 500 kg/day.
4. **Mod 3.4**: **40 refill stations**; avg walk 75m; LP gap 1.31%.
5. **Mod 3.5**: Budget-constrained Pareto frontier; balanced point at Rs. 83.9L.

Unique Contributions (USP)

- **Full geospatial pipeline**: Google Earth KML $\rightarrow$ UTM grid $\rightarrow$ OSM network $\rightarrow$ Dijkstra D_{ij}
- Real walking distances (not Euclidean) — **unique in the class**
- Spatial gravity-based footfall estimation (10 campus venues)
- **3-type waste & vehicle segregation** with LP relaxation bounds

Thank You

"Just like Mother Nature, we optimize continuously — learning, adapting, and improving."

Sustainable Rendezvous Project
Dept. of Chemical Engineering, IIT Delhi